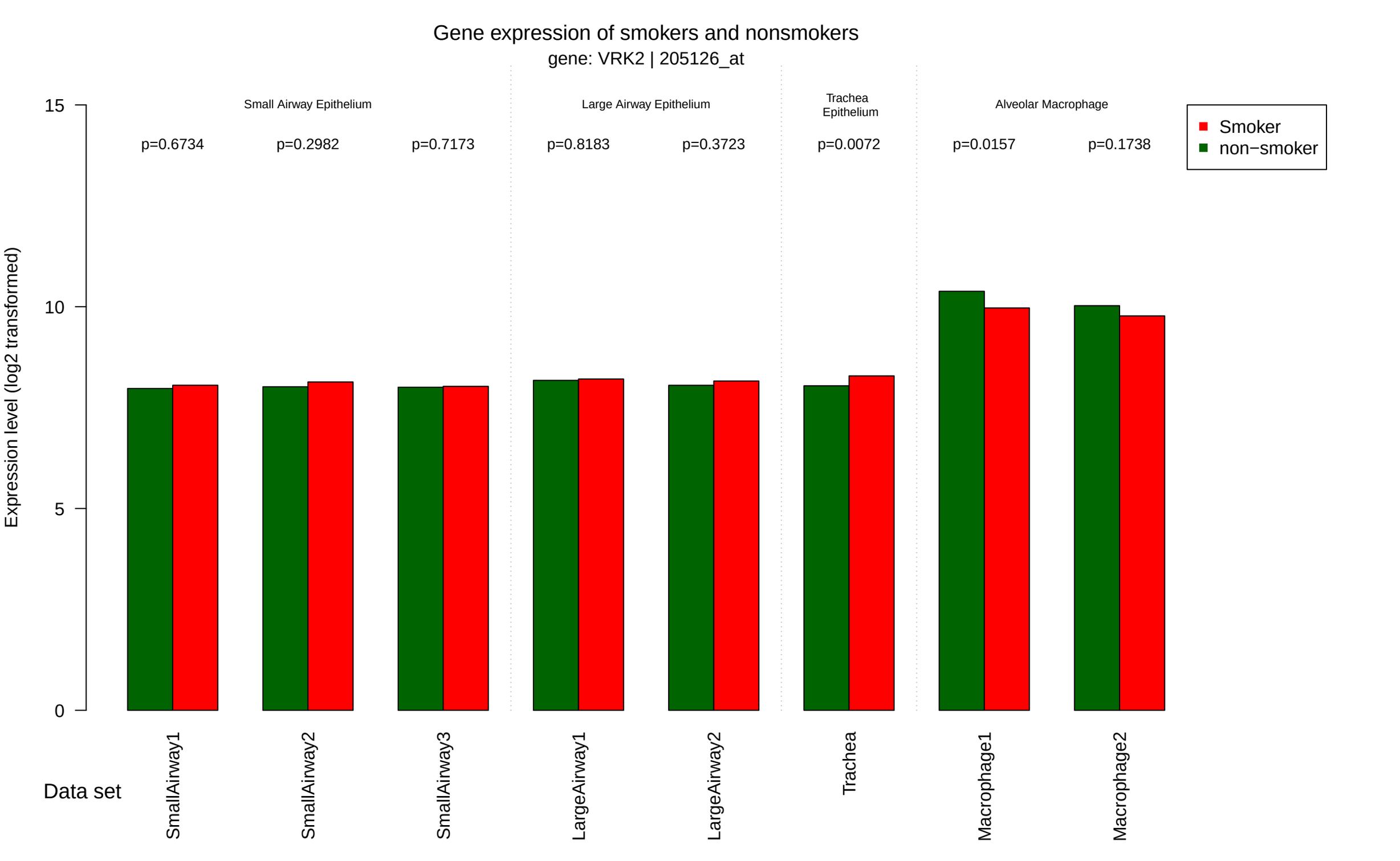
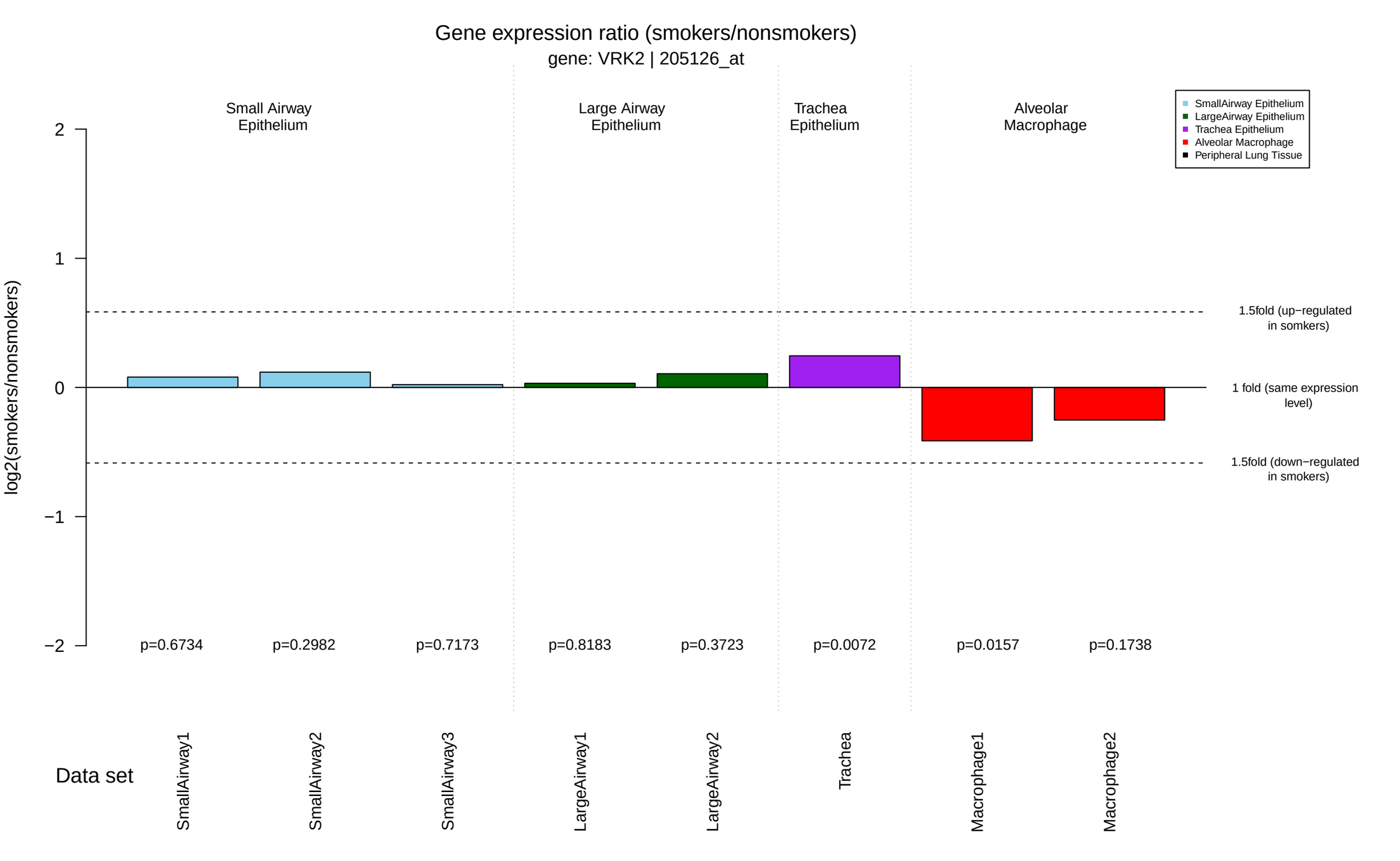
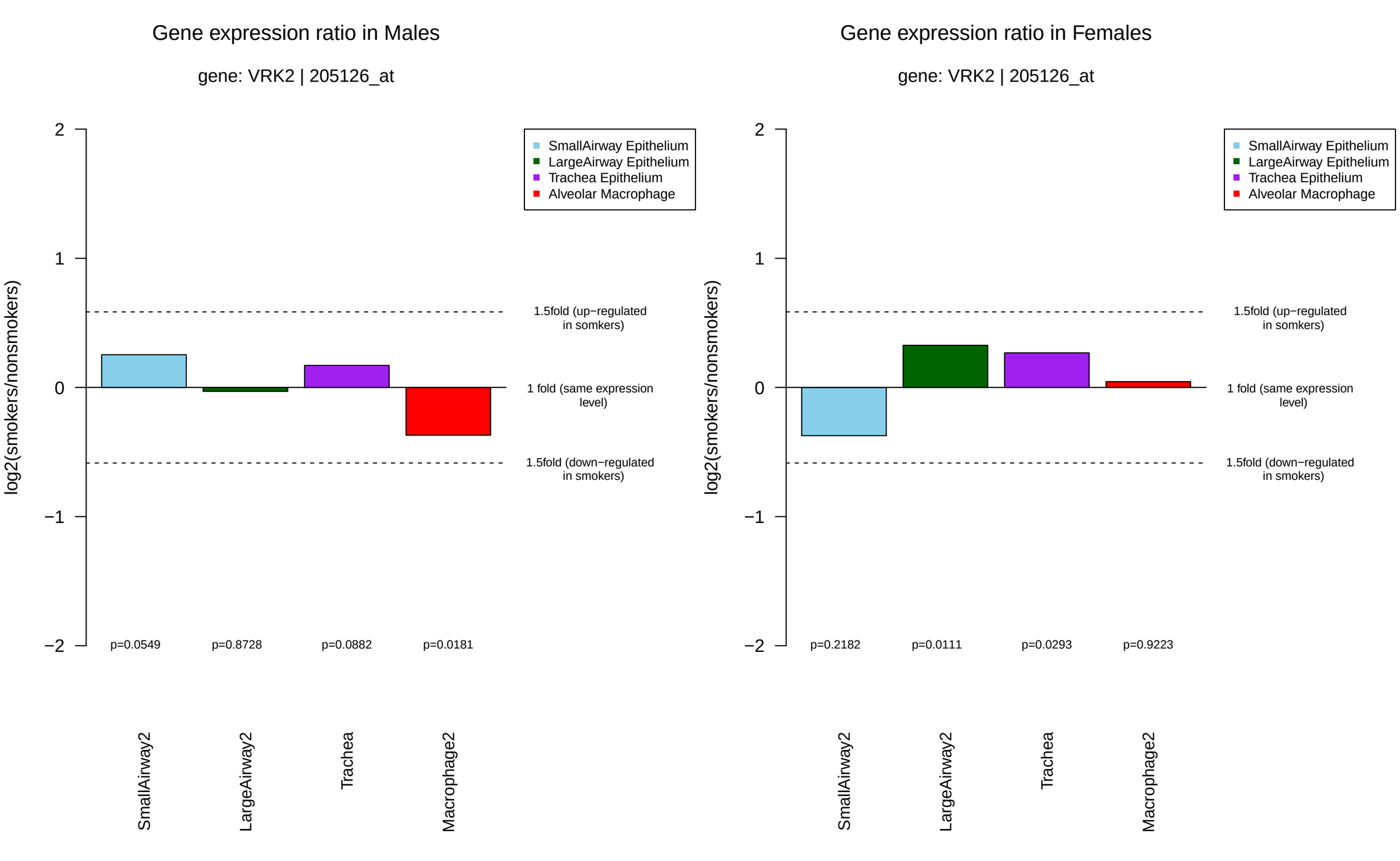
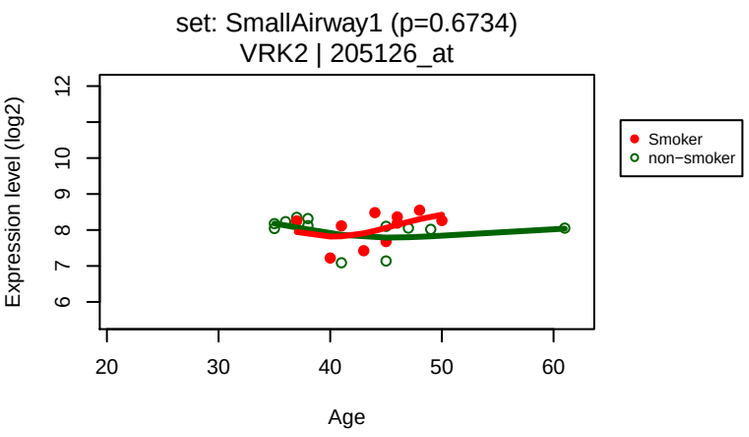
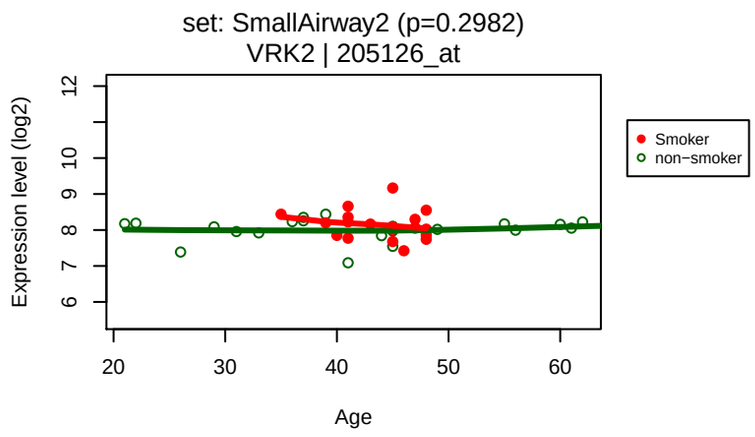
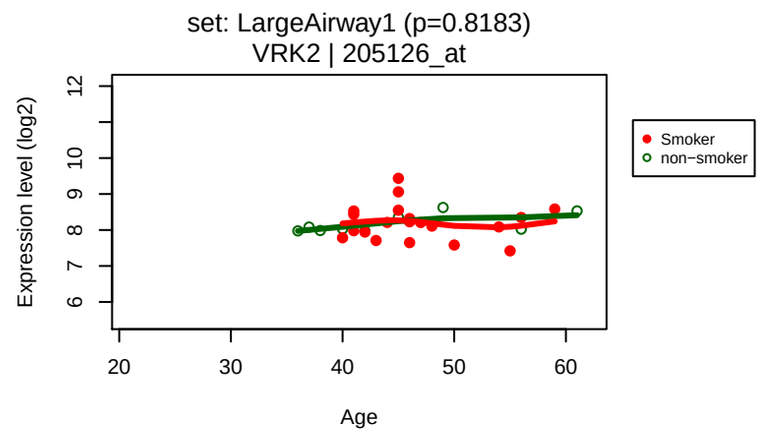

Gene expression of smokers and nonsmokers
gene: VRK2 | 205126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6734
p=0.2982
p=0.7173
p=0.8183
p=0.3723
p=0.0072
p=0.0157
p=0.1738
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

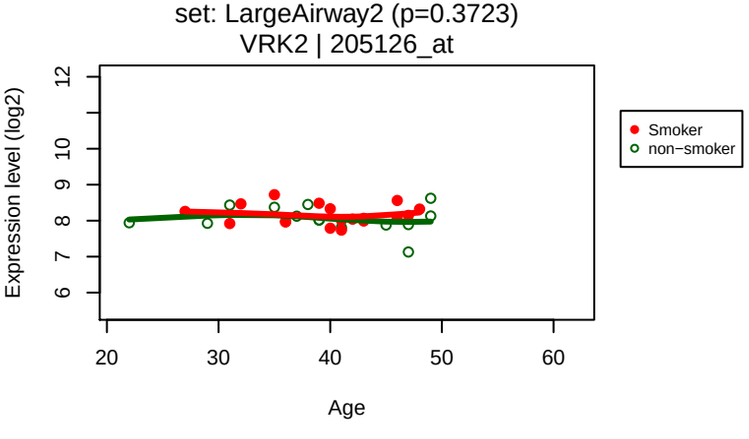

Gene expression ratio (smokers/nonsmokers)
gene: VRK2 | 205126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6734
p=0.2982
p=0.7173
p=0.8183
p=0.3723
p=0.0072
p=0.0157
p=0.1738
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

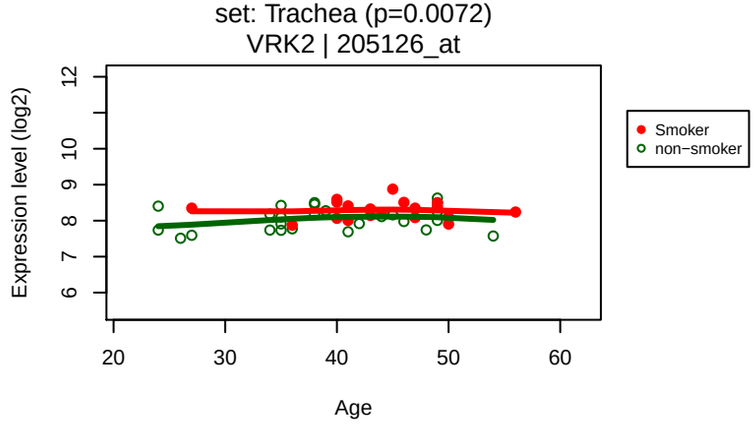

Gene expression ratio in Males
gene: VRK2 | 205126_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0549
p=0.8728
p=0.0882
p=0.0181
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VRK2 | 205126_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2182
p=0.0111
p=0.0293
p=0.9223
SmallAirway2
LargeAirway2
Trachea
Macrophage2

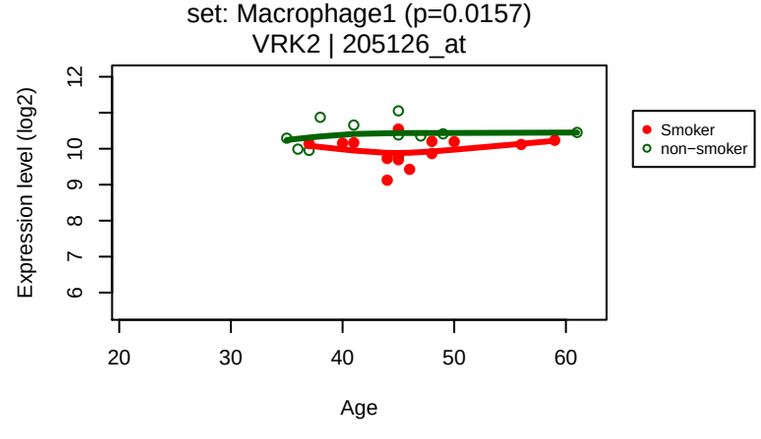

set: SmallAirway1 (p=0.6734)
VRK2 | 205126_at
Expression level (log2)
Age
Smoker
non−smoker

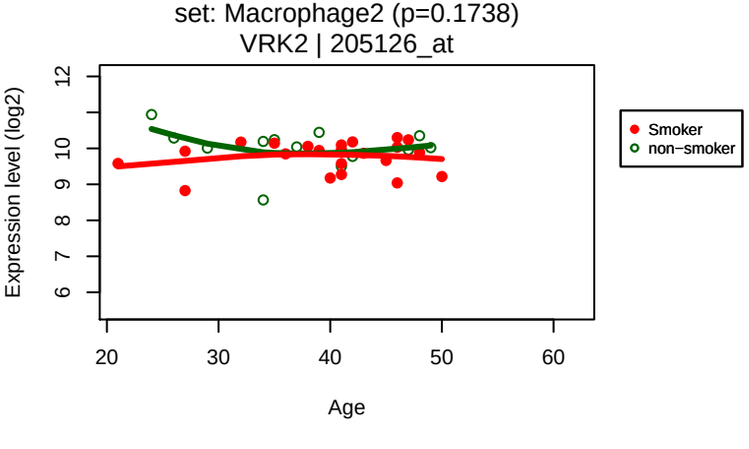

set: SmallAirway2 (p=0.2982)
VRK2 | 205126_at
Expression level (log2)
Age
Smoker
non−smoker

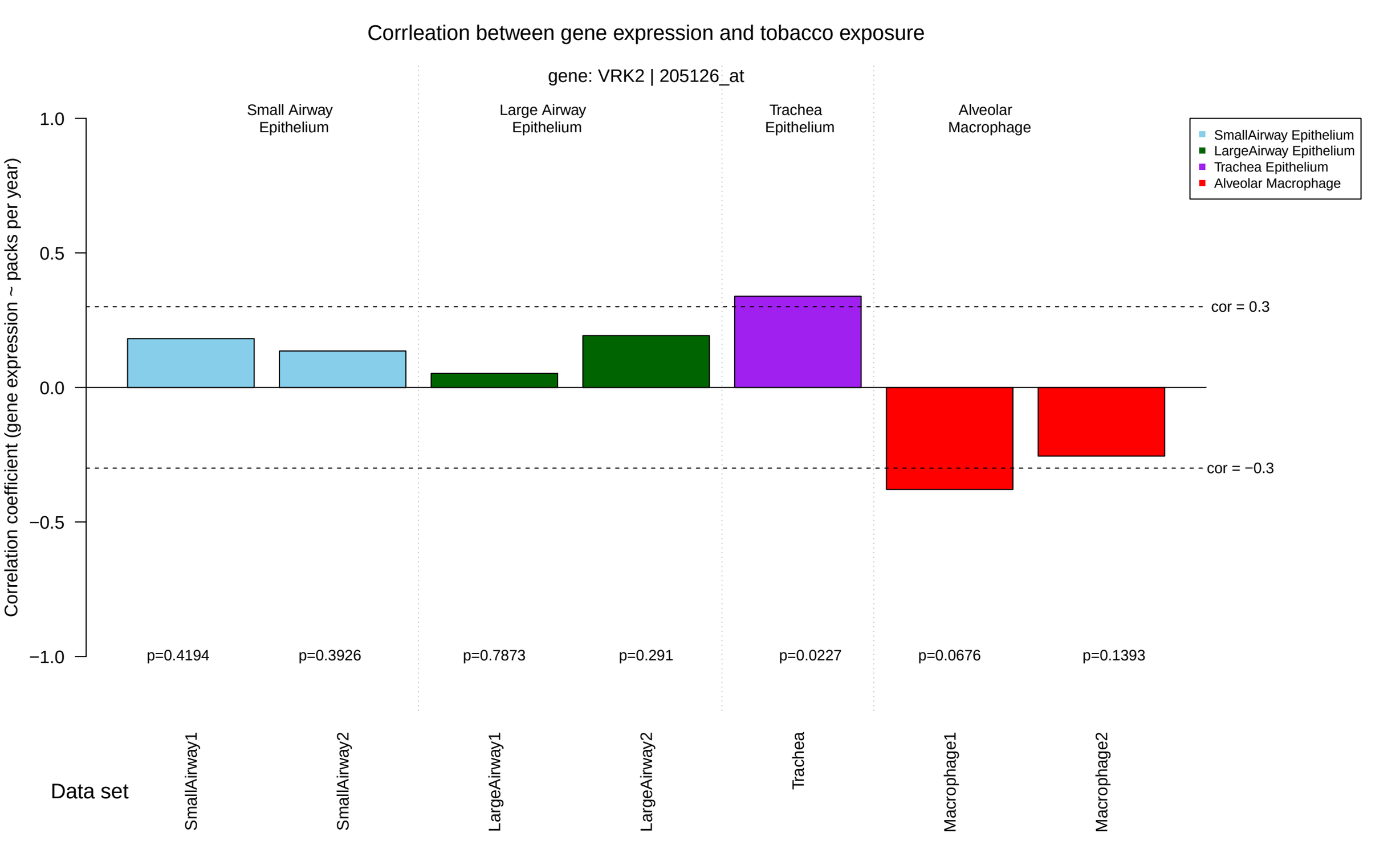

set: LargeAirway1 (p=0.8183)
VRK2 | 205126_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3723)
VRK2 | 205126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0072)
VRK2 | 205126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0157)
VRK2 | 205126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1738)
VRK2 | 205126_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VRK2 | 205126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4194
p=0.3926
p=0.7873
p=0.291
p=0.0227
p=0.0676
p=0.1393
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2